Self Hosted Ghost

Data flow diagram sheets | Revision 1.0 | Sheet 1 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF1 Boundary TB2

EE1 Staff user including Contributor Author Editor; EE2 Owner or Administrator

Credentials verification codes and Admin interactions; local user input →

P1 Admin browser application

DF2 Boundary none

P1 Admin browser application

Open edit save or publish UI; browser internal calls →

P2 Post editor including session save engine and publish UI

DF3 Boundary TB1,TB2

P1 Admin browser application; P2 Post editor including session save engine and publish UI

Staff sign in save publish and image requests with cookie; HTTPS →

P3 Nginx reverse proxy

DF4 Boundary TB1,TB2

EE5 Server side publishing script

Publishing API requests with signed JWT; HTTPS →

P3 Nginx reverse proxy

DF5 Boundary none

P3 Nginx reverse proxy

Forward request and response; loopback HTTP; provider events use P7 →

P4,P5,P6,P7 | See component inventory

DF6 Boundary TB3,TB6

P4 Admin API including staff authentication validation authorization and serialization

Posts roles sessions members settings integration keys; MySQL private TCP →

DS1 MySQL core data

DF7 Boundary none

P4 Admin API including staff authentication validation authorization and serialization

Validated image upload or deletion; in process →

P8 S3 image storage adapter

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.

Self Hosted Ghost

Data flow diagram sheets | Revision 1.0 | Sheet 2 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF8 Boundary TB4

P8 S3 image storage adapter → Authenticated object upload and delete; HTTPS → DS2 S3 public image bucket

DF9 Boundary TB1

EE3 Free member; EE4 Anonymous reader → Public content requests and member login or link redemption; HTTPS → P3 Nginx reverse proxy

DF10 Boundary TB3,TB6

P5 Public site and Content API; P6 Member authentication → Published content member login state and sessions; MySQL private TCP → DS1 MySQL core data

DF11 Boundary none

P4 Admin API including staff authentication validation authorization and serialization → Immediate publish or scheduled publication state; in process with DS1 persistence → P9 Publication scheduler

DF12 Boundary none

P9,P6,P4 | See component inventory → Newsletter jobs member links or staff verification mail; in process → P7 Email delivery and provider event handling

DF13 Boundary TB5

P7 Email delivery and provider event handling → Newsletter API over HTTPS; transactional email over SMTP with TLS → EE6 Mailgun email provider

DF14 Boundary TB7

EE6 Mailgun email provider → Messages containing newsletters or authentication links and codes; email transport → EE7 Recipient mailbox service

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.

Self Hosted Ghost

Data flow diagram sheets | Revision 1.0 | Sheet 3 of 3

Every flow is drawn once. Repeated IDs denote the same element. Arrows include associated responses.
Boundary labels identify crossed trust boundaries defined in the companion package.

External entity | Process | Data store | Dashed frame = flow grouping

DF15 Boundary TB1,TB5

EE6 Mailgun email provider → Delivery and engagement event callbacks; HTTPS → P3 Nginx reverse proxy

DF16 Boundary TB3

P7 Email delivery and provider event handling → Recipient lists and delivery events; MySQL private TCP → DS1 MySQL core data

DF17 Boundary TB4

EE3 Free member; EE4 Anonymous reader → Public image reads; HTTPS without Ghost authorization → DS2 S3 public image bucket

DF18 Boundary none

DS3 Server configuration file → Database email and storage credentials loaded locally → P4,P7,P8 | See component inventory

DF19 Boundary none

P3,P4,P6,P7 | See component inventory → Access authentication and application events; local writes → DS4 Local application and proxy logs

P = process EE = external entity DS = data store TB = trust boundary DF = data flow

Logical view; process boxes do not imply isolation. Full scope and controls are specified in the package.